

TO THANH TAM

Backend Developer | Distributed Systems & Analytics Engineering

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PROFESSIONAL SUMMARY

Backend Engineer & Tech Lead with 4+ years of experience specializing in Event-Driven Microservices, CQRS Architecture, and High-Throughput Streaming Data Pipelines. Proven track record of leading the technical architecture and development for Media Hub (Social Media Analytics Platform) at VNPT-Media and contributing as a Core Backend Developer to large-scale enterprise ERP/CRM systems at FPT IS.

- **Leadership & Core Tech:** Technical Leadership, Node.js (NestJS), Java (Spring Boot), CQRS, Kafka (CDC/DLQ), ClickHouse, Redis BullMQ, Saga Orchestration.
- **Soft Skills:** Project Leadership, technical presentation, cross-functional collaboration, direct solution negotiation with Client IT Directors.

TECHNICAL SKILLS

- **Languages & Frameworks:** Node.js (NestJS), Java (Spring Boot), .NET Framework, SAP ABAP, Angular, SQL.
- **Architecture & Patterns:** Microservices, CQRS (Read/Write Separation), Event-Driven Architecture, Saga Orchestration, HLD.
- **Databases & Storage:** Clickhouse (OLAP/Materialized Views), MongoDB (Replica Set), SQL Server, Redis.
- **Streaming & Messaging:** Apache Kafka (DLQ Fallback/Batching), RabbitMQ, Dagster/Airflow, Redis BullMQ, Fluent-ffmpeg.
- **Methodology & Tools:** Agile/Scrum, Data Modeling (BCNF), Clean Code, RESTful APIs, OAuth2/JWT, Docker.

WORK EXPERIENCE

VNPT-Media | Backend Tech Lead / Software Engineer

Jun 2026 – Present

Key Projects: Media Hub Platform (Social Media Analytics) & Television Analytics Platform

- **Technical Leadership (Media Hub):** Serving as Project Tech Lead for the Media Hub Social Analytics & Publishing Platform; driving technical architecture, task distribution, and system performance standards.
- **CQRS & High-Availability Architecture:** Designed High-Level Architecture (HLD) applying CQRS pattern to strictly separate Read/Write workloads for high-concurrency analytics systems.
- **Multi-Channel Social Campaign Engine:** Architected NestJS APIs for multi-channel content publishing (Reels, Shorts, Link Posts) supporting scheduling, modification, cancellation, and fast retry logic.
- **Streaming Data Pipeline (Kafka -> ClickHouse):** Developed Node.js Ingestion Gateways to collect high-volume logs, pushed events to Kafka topics, and implemented Consumer Workers with batching strategy for ClickHouse ingestion.
- **Fault Tolerant & DLQ Mechanism:** Built Dead Letter Queue (DLQ) fallback mechanisms to catch ingestion errors during database downtime, eliminating log loss.

- **ClickHouse & Analytics APIs Optimization:** Leveraged ClickHouse Materialized Views to auto-parse JSON into flat schemas; built 6 real-time Analytics APIs delivering sub-second response times.
- **Async Scheduling & Pre-processing:** Orchestrated async publishing and token control via Redis BullMQ and Cronjob Scheduler; utilized fluent-ffmpeg to pre-process video metadata (duration, aspect ratio) for platform recommendation.
- **Referrer Tracking & Unified Analytics:** Developed APIs to track traffic logs and extract attribution sources; designed ClickHouse schemas and built Unified Reporting APIs.

FPT IS | Backend Developer

May 2022 - March 2026

Key Project: ERP-DXG Implementation (Client: Dat Xanh Group - 150B VND)

- **Core Backend Developer:** Core backend engineer developing the Central CRM Module, acting as the integration hub for the enterprise satellite ecosystem.
- **Architecture & Security:** Engineered secure NestJS lifecycle (Auth Guard -> Data Isolation -> Exception Filter -> DTO Validation).
- **Distributed Transactions:** Implemented Saga Orchestrator via RabbitMQ to maintain data integrity across internal services within the CRM ecosystem.
- **Satellite Integration:** Orchestrated seamless integration between Core CRM and satellite systems including SAP, iHRP, CONs, ADMs, and eApp.
- **Real-time Analytics:** Architected Kafka (CDC) streams from MongoDB to Clickhouse; managed workflows using Dagster.
- **Mentorship:** Trusted to mentor interns from top universities, standardizing Clean Code and professional Git workflows.

E-Hiring & E-Learning Systems

- **Metadata-driven UI:** Built a MongoDB Metadata-driven mechanism enabling UI configuration changes without code deployment.
- **Integration & Reporting:** Orchestrated category data sync between iHRP and E-learning via .NET Framework; engineered high-performance matrix reports using Advanced SQL Cursors.

SAP ABAP Projects (Acecook & Elanco)

- Developed SD/MM/WM reports with QR-code inventory integration (Acecook); decoded FI/CO logic via Reverse Engineering (Elanco).

ACADEMIC PROJECTS & RESEARCH

- **Mobile Timetable & QR-Attendance System:** Awarded 3rd Prize in Scientific Research (2021-2022); offered full-time contract by TMA Solutions upon defense.
- **Vaccination Management System:** Designed a BCNF-compliant database with automated business rules via Database Triggers.

EDUCATION & CERTIFICATIONS

- Engineer in Software Engineering | HUTECH University (GPA: 3.5 / 4.0)
- Agile & Scrum Professional Certificate | DEK Technologies (2021)
- Internal SAP/ABAP Professional Training | FPT IS (2022)